

Project Design Phase

Problem – Solution Fit Template

Date	25 June 2026
Team ID	
Project Name	Smart Agricultural Production Optimization Engine (OptiCrop)
Maximum Marks	2 Marks

PROBLEM – SOLUTION FIT CANVAS – OPTICROP

Smart Agricultural Production Optimization Engine

1 CUSTOMER SEGMENT(S)  - Farmers seeking guidance on crop selection. - Agricultural researchers analyzing crop and soil data. - Agriculture officers supporting farmers with recommendations.	6 CUSTOMER CONSTRAINTS  - Limited agricultural expertise. - Limited access to professional guidance. - Time required for manual decision-making. - Difficulty analyzing multiple environmental parameters. - Financial risks associated with incorrect crop selection.	5 AVAILABLE SOLUTIONS  - Consulting experienced farmers. - Seeking advice from agriculture officers. - Traditional farming knowledge. - Internet searches and agricultural articles. - Manual analysis of soil reports.
2 JOBS-TO-BE-DONE / PROBLEMS  - Select the most suitable crop based on soil and weather conditions. - Verify whether a specific crop is suitable for cultivation. - Analyze agricultural data to understand crop suitability patterns. - Reduce crop failure and improve agricultural productivity.	9 PROBLEM ROOT CAUSE  - Crop selection depends on several interrelated parameters such as soil nutrients, temperature, humidity, pH, and rainfall. - Farmers often lack an easy-to-use data-driven decision support system. - Manual analysis is time-consuming and may lead to inaccurate decisions. - Agricultural knowledge is not always readily available when required.	7 BEHAVIOUR  - Consult local farmers and agriculture experts. - Review previous farming experiences. - Perform soil testing. - Compare multiple crop options. - Search online for agricultural recommendations before making a decision.
3 TRIGGERS  - Beginning of a new cultivation season. - Soil testing has been completed. - Uncertainty about which crop to cultivate. - Unexpected weather or environmental changes. - Desire to maximize crop yield and profit.	10 OUR SOLUTION  Develop the Smart Agricultural Production Optimization Engine (OptiCrop), a machine learning-based decision support system that analyzes soil nutrients and environmental parameters to recommend the most suitable crop. The system also evaluates crop suitability and supports agricultural dataset analysis through a simple Flask-based web application, enabling farmers and researchers to make faster, more accurate, and data-driven decisions.	8 CHANNELS OF BEHAVIOUR 8.1 ONLINE - OptiCrop Web Application - Agricultural websites - Search engines - Mobile devices - Educational resources 8.2 OFFLINE - Agriculture officers - Local farming communities - Soil testing laboratories - Farmer meetings - Agricultural training programs
4 EMOTIONS (BEFORE / AFTER)  BEFORE - Confused - Uncertain  AFTER - Confident - Informed ----->		